

Harshit Gupta

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SUMMARY

GCP-native Data Engineer with 3.9 years of experience building and optimizing large-scale data pipelines using **PySpark**, **BigQuery**, **Dataproc**, and **Apache Airflow** on Google Cloud Platform. Proven track record at American Express delivering robust ETL workflows, Spark performance tuning, and secure data transfer solutions in a high-stakes enterprise environment.

SKILLS

Languages and Frameworks: Python3, Apache Spark, PySpark, SQL, Apache Airflow, BigQuery, Google Cloud Platform, PyTest.
Data Engineering and Cloud: GCP – BigQuery, Dataproc, Cloud Composer, Pub/Sub, Dataflow, Databricks, Snowflake.
AI Assisted Development: GitHub Copilot, Claude Code, ChatGPT, Google Agent Development Kit, Model Context Protocol.
Secondary Tech Stack: HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Bootstrap.
Additional Skills: Data Structures and Algorithms, Git, GitHub, Agile Methodology (Jira, Rally).

CERTIFICATIONS

Google Cloud Platform – Data Engineer, Big Data

WORK EXPERIENCE

Infosys Ltd

Bangalore, India

Client: American Express – Engineer II

August 2022 – Present

- Engineered robust **PySpark** ETL pipelines on **Dataproc** to process and migrate complex multi-partition datasets from **BigQuery** to enterprise mainframe systems utilizing **complex analytical queries** for downstream transformations.
- Designed and maintained end-to-end enterprise data solutions by architecting data pipelines across the full **GCP** data ecosystem, seamlessly integrating **BigQuery** for warehousing, **Dataproc** for Spark processing, **Cloud Composer** for orchestration, **Cloud Storage** for data lake, **Pub/Sub** for messaging, **Dataflow** for ingestion, and **Cloud Functions** for event triggers.
- Architected **BigQuery** storage patterns using **table partitioning** and **clustering** to minimize data scans, reduce slot consumption, and lower query costs across large enterprise datasets.
- Automated and monitored **40+ Airflow DAGs** in **GCP Cloud Composer** processing **~1 TB of data daily**; managed complex **task dependencies** and integrated **Cloud Pub/Sub** for event-driven pipeline triggering.
- Tuned **Spark** performance on **Dataproc** via **broadcast joins**, **data caching**, **skew mitigation**, **dynamic allocation**, and **ephemeral clusters**, **reducing execution time by ~30%**.
- Gained hands-on exposure to **Dataflow (Apache Beam)** and **Cloud Functions** for batch data ingestion patterns and lightweight event-triggered processing on GCP.
- Developed **Python**-based SFTP scripts with **PGP Encryption** to securely transfer files from **GCS** to mainframe; wrote **Pytest** unit tests and **BDD** (Gherkin) scenarios to ensure pipeline reliability.
- Leveraged **Agile** methodologies (Rally, Jira) to collaborate with cross-functional teams, manage sprint deliverables, and align pipeline development with business requirements.

PROJECTS

Data Processing Pipeline | *Python3, PySpark, Apache Spark, SQLite*

Website | GitHub

- Demonstrated a simple data processing pipeline designed to clean, transform, and analyze data.
- Optimized join operations using **broadcast joins** to efficiently combine resulting Spark DataFrames.
- Validated records by identifying suspect records and separated good records from bad ones.
- Created an appropriate data model in an **SQLite** database and ingested the cleaned data into it.

Skills Learned: Data extraction, Data handling and manipulation, Data ingestion

YelpCamp | *MongoDB, Express.js, EJS, Node.js, Passport.js, Bootstrap*

Website | GitHub

- Developed a full-stack campground review site with user accounts, campground listings, and review posting.
- Built complete **CRUD** functionality for campgrounds and reviews, adhering to **RESTful API** principles.

Skills Learned: Full-Stack Development, CRUD operations, RESTful API Principles

DigiMenu | *HTML, CSS, JavaScript, MongoDB, Node.js, Express.js, Passport.js*

Website | GitHub

- Built a digital menu application allowing users to browse and view available items and categories.
- Implemented **authentication and authorization** using **Passport.js** with session-based login.

Skills Learned: CRUD operations, RESTful routing, Authentication and Authorization

EDUCATION

Visvesvaraya Technological University

Bangalore, Karnataka, IN

B.E. in Electronics and Communication | Overall: 8.8 CGPA

August 2018 – July 2022